

# AQUÍ VA EL TÍTULO DEL ARTÍCULO

JUAN CARLOS MONTER CORTÉS AND ÁNGEL ZALDÍVAR CORICHI

ABSTRACT. Aquí va el resumen

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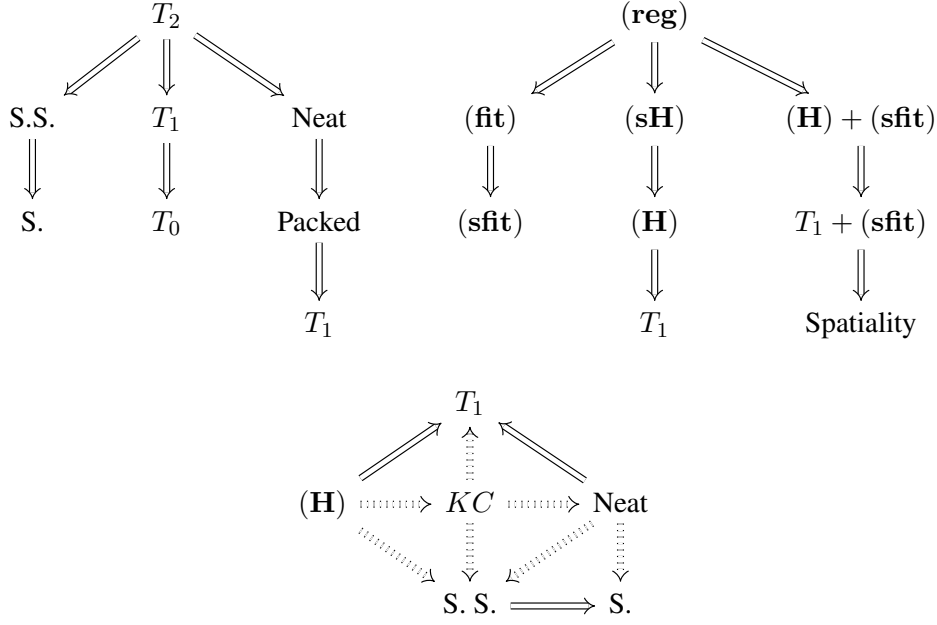
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## 1. INTRODUCTION

Completar la introducción.

[6, 2]

DONE: hay que modificar el formato bastante, hay muchas separaciones en las demostraciones eso se ve raro tambien te sugiero que agreges un paper que te guste y veas su estructura de como escriben las cosas



## 2. PRELIMINARIES

DONE: Combinar secciones 2 y 3

**2.1. Frames, nuclei, and open filters.** We begin by recalling some basic facts and terminology concerning frames, nuclei, and open filters. We refer to [3, 5, 11, 13] for general background on frames and point-free topology. A frame is a complete lattice  $A$  satisfying the infinite distributive law

$$a \wedge \bigvee X = \bigvee \{a \wedge s \mid s \in X\}$$

for every  $a \in A$  and  $X \subseteq A$ . A frame homomorphism  $f: A \rightarrow B$  is a map preserving finite meets and arbitrary joins. Frames and frame homomorphisms form the category  $\mathbf{Frm}$ .

As usual, the distributivity law determines a Heyting implication  $\succ$  on  $A$ , characterized by

$$x \wedge a \leq b \iff x \leq (a \succ b).$$

In particular, every frame homomorphism  $f: A \rightarrow B$ , being a map preserving arbitrary joins, has a right adjoint  $f_*: B \rightarrow A$ , characterized by

$$f(a) \leq b \iff a \leq f_*(b).$$

A nucleus on a frame  $A$  is a map  $j: A \rightarrow A$  satisfying

$$x \leq j(x), \quad j(j(x)) = j(x), \quad j(x \wedge y) = j(x) \wedge j(y)$$

for all  $x, y \in A$ . The set

$$A_j = \{x \in A \mid (x) = x\}$$

of fixed points of  $j$  is a frame, where arbitrary joins are given by

$$\bigvee_{A_j} S = j \left( \bigvee_A S \right).$$

Moreover, the map

$$j^*: A \longrightarrow A_j, \quad j^*(x) = j(x),$$

is a surjective frame homomorphism. Conversely, every quotient of  $A$  arises in this way. We shall therefore identify nuclei on  $A$  with the corresponding quotients  $A_j$ . Three particular families of nuclei will be used throughout the paper. For each  $a \in A$ , let

$$u_a(x) = a \vee x, \quad v_a(x) = a \rightarrow x, \quad w_a(x) = (x \rightarrow a) \rightarrow a.$$

The nuclei  $u_a$  and  $v_a$  determine, respectively, the closed and open quotient associated with  $a$ , while the  $w_a$ 's will play an important role in the study of compact quotients below.

An element  $p \in A$  is  $\wedge$ -irreducible if  $p \neq 1$  and

$$x \wedge y \leq p \implies x \leq p \text{ or } y \leq p.$$

We denote by  $\text{pt}(A)$  the set of all  $\wedge$ -irreducible elements of  $A$ . These are in bijective correspondence with the completely prime filters of  $A$ : to each  $p \in \text{pt}(A)$  corresponds the completely prime filter

$$F_p = \{a \in A \mid a \not\leq p\}.$$

Conversely, if  $P$  is a completely prime filter, then

$$p_P = \bigvee (A \setminus P)$$

is  $\wedge$ -irreducible and  $P = F_{p_P}$ .

The point space of  $A$  is the set  $\text{pt}(A)$  equipped with the topology generated by the sets

$$\mathcal{U}(a) = \{p \in \text{pt}(A) \mid a \not\leq p\}, \quad a \in A.$$

In fact, the assignment

$$\mathcal{U}: A \longrightarrow \mathcal{O}(\text{pt}(A)), \quad a \longmapsto \mathcal{U}(a),$$

is a surjective frame homomorphism.

The map  $\mathcal{U}$  is usually referred to as the spatial reflection of  $A$ . A frame  $A$  is said to be spatial if  $\mathcal{U}$  is an isomorphism, that is, if

$$A \cong \mathcal{O}(\text{pt}(A)).$$

Equivalently,  $A$  is spatial whenever its points separate its elements: for  $a, b \in A$ ,

$$a \not\leq b \implies \text{there exists } p \in \text{pt}(A) \text{ such that } b \leq p \text{ and } a \not\leq p.$$

A filter  $F$  on a frame  $A$  is said to be open if, for every directed subset  $D \subseteq A$ ,

$$\bigvee D \in F \implies D \cap F \neq \emptyset.$$

Thus, open filters are precisely those filters which are open with respect to the Scott topology on  $A$ . We denote by  $A^\wedge$  the collection of all open filters on  $A$ .

Open filters can be transferred along frame morphisms. Let

$$f = f^*: A \longrightarrow B$$

be a frame morphism with right adjoint  $f_*: B \rightarrow A$ . If  $F$  and  $G$  are filters on  $A$  and  $B$ , respectively, we define filters  $f^*F$  on  $B$  and  $f_*G$  on  $A$  by

$$(1) \quad b \in f^*F \iff f_*(b) \in F, \quad a \in f_*G \iff f(a) \in G.$$

**Proposition 2.1.** *Let  $f = f^*: A \rightarrow B$  be a frame morphism. If  $G \in B^\wedge$ , then  $f_*G \in A^\wedge$ .*

*Proof.* Clearly,  $f_*G$  is a filter on  $A$ . Let  $X \subseteq A$  be directed and suppose that

$$\bigvee X \in f_*G.$$

Then  $f[X]$  is directed and, since  $f$  preserves arbitrary joins,

$$\bigvee f[X] = f\left(\bigvee X\right) \in G.$$

Since  $G$  is open, there exists  $x \in X$  such that  $f(x) \in G$ . Hence  $x \in f_*G$ , and therefore  $f_*G \in A^\wedge$ .  $\square$

Every nucleus  $j$  on  $A$  determines a filter

$$\nabla(j) = \{a \in A \mid j(a) = 1\},$$

called the admissible filter of  $j$ . We say that  $j$  admits  $a \in A$  whenever  $j(a) = 1$ .

Two nuclei  $j, k$  are said to be companions whenever

$$\nabla(j) = \nabla(k).$$

Each class of companion nuclei has a least element. These least elements are called fitted nuclei.

Given a filter  $F$  on  $A$ , consider the pre-nucleus

$$f_F = \bigvee_{a \in F} v_a,$$

where the join is taken pointwise. Explicitly,

$$f_F(x) = \bigvee_{a \in F} (a \succ x).$$

Its transfinite iteration yields the least nucleus above  $f_F$ , which we denote by

$$v_F = f_F^\infty.$$

Equivalently,

$$v_F = \bigvee_{a \in F} v_a,$$

where the join on the right-hand side is computed in the frame  $NA$  of nuclei on  $A$ .

The nucleus  $v_F$  is the least nucleus on  $A$  admitting every element of  $F$ . Consequently,

$$F \subseteq \nabla(v_F).$$

Moreover, if  $F$  is open, then

$$\nabla(v_F) = F.$$

Hence, the assignment

$$F \longmapsto v_F$$

gives a one-to-one correspondence between open filters and fitted nuclei whose admissible filters are open.

Open filters will play a central role throughout the paper. Besides determining fitted nuclei, they provide the point-free counterparts of compact saturated subsets through the Hofmann–Mislove correspondence, which we recall next.

Let  $S = \text{pt}(A)$ , endowed with its natural topology. We denote by  $\mathcal{Q}(S)$  the collection of compact saturated subsets of  $S$ . The Hofmann–Mislove theorem establishes a bijective correspondence

$$A^\wedge \longleftrightarrow \mathcal{Q}(\text{pt}(A))$$

between open filters on  $A$  and compact saturated subsets of its point space.

More explicitly, if  $F \in A^\wedge$ , its corresponding compact saturated subset is

$$Q_F = \{p \in \text{pt}(A) \mid (\forall a \in F) a \not\leq p\},$$

whereas, for  $Q \in \mathcal{Q}(\text{pt}(A))$ , the corresponding open filter is

$$F_Q = \{a \in A \mid (\forall p \in Q) a \not\leq p\}.$$

Thus,

$$a \in F_Q \iff Q \subseteq \mathcal{U}(a).$$

When  $A = \mathcal{O}(S)$  for a sober space  $S$ , this correspondence reduces to the classical Hofmann–Mislove theorem: the compact saturated subsets  $Q \subseteq S$  correspond bijectively to the open filters of  $\mathcal{O}(S)$  through

$$Q \longmapsto \nabla(Q) = \{U \in \mathcal{O}(S) : Q \subseteq U\}.$$

Whenever  $F \in A^\wedge$ , we shall denote by  $Q$  its corresponding compact saturated subset of  $\text{pt}(A)$ , unless it is necessary to emphasize the dependence on  $F$ . We shall refer to a pair  $(F, Q)$ , where  $F \in A^\wedge$  and  $Q \in \mathcal{Q}(\text{pt}(A))$  correspond under the above bijection, as an HM pair.

The Hofmann–Mislove correspondence can be extended to include fitted nuclei. More precisely, the Hofmann–Mislove–Johnstone theorem establishes a correspondence between open filters, compact saturated subsets, and the associated fitted nuclei.

The preceding correspondence allows us to regard open filters as point-free representatives of compact saturated subsets. In the next subsection we shall see that they also determine compact quotients of  $A$ , and we shall study the nuclei associated with such quotients.

**2.2. Compact Quotients and Admissibility Intervals.** We now turn our attention to compact quotients. Recall that every nucleus  $j$  on a frame  $A$  determines a quotient  $A_j$ , and that its admissible filter is given by

$$\nabla(j) = \{a \in A : j(a) = 1\}.$$

The following characterization of compact quotients is due to Sexton [9, Lemma 5.17].

**Proposition 2.2.** *Let  $j$  be a nucleus on a frame  $A$ . Then the quotient  $A_j$  is compact if and only if  $\nabla(j)$  is an open filter.*

In particular, for every  $F \in A^\wedge$ , the fitted nucleus  $v_F$  determines a compact quotient  $A_{v_F}$ . Conversely, every compact quotient  $A_j$  determines the open filter  $\nabla(j) \in A^\wedge$ . Notice, however, that a compact quotient is not, in general, uniquely determined by its admissible filter. Thus, for a fixed  $F \in A^\wedge$ , it is natural to consider all nuclei  $j$  satisfying

$$\nabla(j) = F.$$

As recalled above,  $v_F$  is the least such nucleus. We shall now describe the nuclei having  $F$  as their admissible filter and the corresponding interval of admissibility.

Let  $F \in A^\wedge$ . Following the construction used in the general form of the H–M theorem, consider

$$M_F = \{m \in A \setminus F : m \text{ is maximal}\}.$$

Associated with  $F$ , define

$$w_F(x) = \bigwedge \{m \in M_F : x \leq m\}, \quad x \in A,$$

where the meet of the empty family is understood to be 1.

If  $\text{pt}(A)$  is  $T_1$ , then every point is maximal and, for the HM pair  $(F, Q)$ , one has

$$M_F = Q.$$

The nucleus  $w_F$  is the largest nucleus having  $F$  as its admissible filter. Hence

$$\nabla(w_F) = F.$$

It follows that the nuclei having  $F$  as their admissible filter form the interval

$$[v_F, w_F] = \{j \in N(A) : v_F \leq j \leq w_F\}.$$

We shall refer to  $[v_F, w_F]$  as the admissibility interval associated with  $F$ .

Evaluating the endpoints at 0, we obtain the interval

$$[v_F(0), w_F(0)] \subseteq A,$$

which will also play a central role in what follows.

We shall also need a particular class of nuclei arising from subspaces. Let  $S$  be a topological space, and denote by  $E^\circ$  the interior of a subset  $E \subseteq S$ . For each  $E \subseteq S$ , consider the map

$$[E] : \mathcal{O}S \longrightarrow \mathcal{O}S, \quad [E](U) = (E \cup U)^\circ.$$

Here, for each  $E \subseteq S$ , we denote by  $E^\circ$  the interior of  $E$  in  $S$ . It is straightforward to verify that  $[E]$  is a nucleus on  $\mathcal{O}S$ . A nucleus  $j$  on  $\mathcal{O}S$  is said to be spatially induced if  $j = [E]$  for some  $E \subseteq S$ .

Spatially induced nuclei provide a natural representative within the admissibility interval associated with the neighbourhood filter of a compact saturated subset.

**Example 2.3.** Let  $Q$  be a compact saturated subset of a topological space  $S$ , and consider its neighbourhood filter

$$G = \nabla(Q) = \{U \in \mathcal{O}S \mid Q \subseteq U\}.$$

By the Hofmann–Mislove theorem,  $G \in (\mathcal{O}S)^\wedge$ . Consider the spatially induced nucleus  $[Q']$ , where  $Q' = S \setminus Q$ . For every  $U \in \mathcal{O}S$ ,

$$[Q'](U) = S \iff (Q' \cup U)^\circ = S \iff Q' \cup U = S \iff Q \subseteq U.$$

Hence

$$\nabla([Q']) = G.$$

Since  $v_G$  and  $w_G$  are, respectively, the least and greatest nuclei having  $G$  as their admissible filter, it follows that

$$v_G \leq [Q'] \leq w_G.$$

and evaluating in  $\emptyset$ ,  $v_G(\emptyset) \subseteq (Q')^\circ \subseteq w_G(\emptyset)$ .

More generally, let  $F \in A^\wedge$ , let  $Q \subseteq \text{pt}(A)$  be its corresponding compact saturated subset, and set  $G = \nabla(Q)$ . We shall call  $(F, Q, G)$  an HM triple.

**Proposition 2.4.** Let  $(F, Q, G)$  be an HM triple. If  $k \in [v_G, w_G]$ , then

$$\mathcal{U}_* \circ k \circ \mathcal{U} \in [v_F, w_F].$$

*Proof.* Since  $k$  is a nucleus on  $\mathcal{O}S$ , the composite

$$\mathcal{U}_* \circ k \circ \mathcal{U}$$

is a nucleus on  $A$ . Thus, it suffices to show that

$$\nabla(\mathcal{U}_* \circ k \circ \mathcal{U}) = F.$$

Indeed, for every  $x \in A$ ,

$$\begin{aligned} x \in F &\iff Q \subseteq \mathcal{U}(x) \\ &\iff \mathcal{U}(x) \in G \\ &\iff k(\mathcal{U}(x)) = S \\ &\iff \mathcal{U}_*(k(\mathcal{U}(x))) = 1 \\ &\iff x \in \nabla(\mathcal{U}_* \circ k \circ \mathcal{U}). \end{aligned}$$

Here, the third equivalence follows from  $\nabla(k) = G$ , while the fourth follows from the adjunction  $\mathcal{U} \dashv \mathcal{U}_*$  and  $\mathcal{U}(1) = S$ . Hence

$$\nabla(\mathcal{U}_* \circ k \circ \mathcal{U}) = F,$$

and therefore

$$\mathcal{U}_* \circ k \circ \mathcal{U} \in [v_F, w_F].$$

□

Proposition 2.4 defines a map

$$\Phi: [v_G, w_G] \longrightarrow [v_F, w_F], \quad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U}.$$

In particular, for every  $k \in [v_G, w_G]$ , we have

$$(2) \quad v_F \leq \Phi(k)\mathcal{U} \leq w_F.$$

We conclude this section by recording the behavior of spatially induced nuclei under this lifting in the spatial case.

**Lemma 2.5.** *Let  $A = \mathcal{O}S$  be spatial and let  $E \subseteq S$ . Identifying  $S$  with  $\text{pt}(\mathcal{O}S)$ , we have*

$$\mathcal{U}_* \circ [E] \circ \mathcal{U} = [E].$$

*Proof.* Since  $A = \mathcal{O}S$  is spatial,

$$\mathcal{U}: \mathcal{O}S \longrightarrow \mathcal{O}(\text{pt}(\mathcal{O}S))$$

is an isomorphism. Let  $y \in \mathcal{O}S$ . Since  $\mathcal{U}$  is surjective, there exists  $x \in \mathcal{O}S$  such that

$$(E \cup \mathcal{U}(y))^\circ = \mathcal{U}(x).$$

Hence

$$\begin{aligned} (\mathcal{U}_* \circ [E] \circ \mathcal{U})(y) &= \mathcal{U}_*((E \cup \mathcal{U}(y))^\circ) \\ &= \mathcal{U}_*(\mathcal{U}(x)) \\ &= x. \end{aligned}$$

On the other hand,

$$\mathcal{U}([E](y)) = (E \cup \mathcal{U}(y))^\circ = \mathcal{U}(x).$$

Since  $\mathcal{U}$  is injective,  $[E](y) = x$ . Therefore

$$(\mathcal{U}_* \circ [E] \circ \mathcal{U})(y) = [E](y)$$

for every  $y \in \mathcal{O}S$ , and consequently

$$\mathcal{U}_* \circ [E] \circ \mathcal{U} = [E].$$

□

### 3. SEPARATION PROPERTIES INDUCED BY COMPACT QUOTIENTS

The notions studied in this section are largely motivated by the work of Sexton and Simmons on point-sensitive and point-free patch constructions. In particular, Sexton introduced the notion of a neat frame in the study of the patch assembly. Although neatness was originally considered in that context, one of its consequences suggests a different point of view that will be central here: for every open filter  $F$ , the fitted nucleus  $v_F$  determines a compact quotient which is also closed.

Our purpose is to study this phenomenon independently of its original role in patch constructions. More precisely, we are interested in identifying those nuclei associated with open filters that determine compact closed quotients, and in understanding what happens when this requirement is imposed not only on the fitted nuclei  $v_F$ , but on other nuclei lying in their corresponding admissibility intervals. This perspective leads naturally to the notions introduced below.

**3.1. Neat frames.** We first recall Sexton's original definition of neatness (it is mentioned in [10, 7, 8]). Besides providing the starting point for the properties studied in this section, this formulation will make explicit the phenomenon concerning compact closed quotients that motivates our approach.

**Definition 3.1.** Let  $A$  be a frame, let  $F \in A^\wedge$  and  $\alpha \in \text{Ord}$ . We say that:

- $F$  is  $\alpha$ -neat, if

$$x \vee d = 1, \quad \text{for every } x \in F,$$

where  $d = f_F^\alpha(0)$ .

- $A$  is  $\alpha$ -neat if every  $F \in A^\wedge$  is  $\alpha$ -neat.
- $A$  is neat if it is  $\alpha$ -neat for some  $\alpha$ .

Let us point out a consequence of the preceding definition that will be relevant throughout this work. Suppose that  $A$  is  $\alpha$ -neat and let  $F \in A^\wedge$ . If

$$d = f_F^\alpha(0),$$

then  $x \vee d = 1$  for every  $x \in F$ . Hence,  $u_d$  admits every element of  $F$ , and therefore

$$f_F \leq u_d.$$

Since  $u_d$  is a nucleus, it follows that

$$v_F = f_F^\infty \leq u_d.$$

On the other hand,  $d = f_F^\alpha(0) \leq f_F^\infty(0) = v_F(0)$ , and hence

$$u_d \leq u_{v_F(0)} \leq v_F.$$

Consequently,

$$v_F = f_F^\infty = u_d.$$

Thus, neatness ensures that the fitted nucleus associated with every open filter is a closed nucleus. Since  $v_F$  determines a compact quotient, every such quotient is simultaneously compact and closed.

Beyond this compact-closed quotient phenomenon, Sexton's study of neatness establishes connections with classical separation properties of topological spaces. This naturally raises the question of whether neatness and the related notions arising from her work admit analogous connections with point-free separation properties. Of particular relevance to our purposes is the spatial characterization of neatness in terms of packed and stacked spaces.

More precisely, Sexton proved that a topological space is neat if and only if it is both packed and stacked ([7, Theorem 6.2]).

**3.2. Stacked and strongly stacked frames.** Motivated by the preceding spatial characterization, we now introduce point-free counterparts of the stacked and strongly stacked conditions.

We first recall the spatial notions introduced by Sexton. Let  $S$  be a topological space, let  $Q \subseteq S$  be compact saturated, and let

$$G = \nabla(Q) \in (\mathcal{O}S)^\wedge$$

be its corresponding open filter. Sexton calls  $S$  stacked if

$$v_G(\emptyset) = (Q')^\circ,$$

for every such pair  $(G, Q)$ , and strongly stacked if

$$v_G = [Q']$$

for every such pair.

We now return to the general setting. Let  $A$  be a frame,  $S = \text{pt}(A)$ , and let  $(F, Q, G)$  be an H–M triple. Recall that the map of Proposition (2.4)

$$\Phi : [v_G, w_G] \longrightarrow [v_F, w_F], \quad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U},$$

provides a natural way of comparing the admissibility interval associated with  $G$  with that associated with  $F$ .

**Proposition 3.2.** *Let  $(F, Q, G)$  be an H–M triple. Then  $\Phi(w_G) = w_F$ , that is,*

$$w_F = \mathcal{U}_* \circ w_G \circ \mathcal{U}.$$

*Proof.* By Proposition 2.4, since  $w_G \in [v_G, w_G]$ , we have

$$\mathcal{U}_* \circ w_G \circ \mathcal{U} \leq w_F.$$

Thus, it remains to prove the reverse inequality.

By adjunction, it is enough to show that

$$\mathcal{U} \circ w_F \leq w_G \circ \mathcal{U}.$$

Let  $x \in A$ . Since  $w_G = [M']$ , we have

$$w_G(\mathcal{U}(x)) = [M'](\mathcal{U}(x)) = (M' \cup \mathcal{U}(x))^\circ.$$

Hence, we need to prove that

$$\mathcal{U}(w_F(x)) \subseteq (M' \cup \mathcal{U}(x))^\circ.$$

Since  $\mathcal{U}(w_F(x))$  is open, it is enough to show that

$$\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x).$$

Suppose, by contradiction, that there exists  $y \in \mathcal{U}(w_F(x))$  such that  $y \notin M'$  and  $y \notin \mathcal{U}(x)$ . Then  $y \in M$  and  $x \leq y$ . On the other hand,  $y \in \mathcal{U}(w_F(x))$  implies  $w_F(x) \not\leq y$ . However,

$$w_F(x) = \bigwedge \{m \in M \mid x \leq m\},$$

and since  $y \in M$  with  $x \leq y$ , it follows that  $w_F(x) \leq y$  a contradiction.

Therefore,  $\mathcal{U}(w_F(x)) \subseteq (M' \cup \mathcal{U}(x))^\circ$ .  $\square$

The preceding proposition shows that  $\Phi$  preserves the upper endpoints of the corresponding admissibility intervals:

$$\Phi(w_G) = w_F.$$

It is therefore natural to ask to what extent the lower endpoints are also preserved. Motivated by Sexton's spatial notions, we consider two possible forms of this

requirement: agreement at the bottom element, and agreement of the nuclei themselves. Since  $[Q']$  is the spatially induced nucleus appearing in Sexton's definitions, its image

$$\Phi([Q']) = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$$

provides the natural candidate for reproducing this behavior on  $A$ . This leads to the following definitions.

**Definition 3.3.** Let  $A$  be a frame. We say that  $A$  is *stacked* if, for every HM pair  $(F, Q)$ ,

$$(U_* \circ [Q'] \circ U)(0) = v_F(0).$$

**Definition 3.4.** Let  $A$  be a frame. We say that  $A$  is *strongly stacked* if, for every HM pair  $(F, Q)$ ,

$$U_* \circ [Q'] \circ U = v_F;$$

Clearly, every strongly stacked frame is stacked.

### 3.3. KC frames.

**Definition 3.5.** Let  $A$  be a frame. We say that  $A$  is *KC* if every compact quotient of  $A$  is closed.

Equivalently,  $A$  is KC if, for every  $j \in NA$  such that

$$\nabla(j) \in A^\wedge,$$

there exists  $a \in A$  such that

$$j = u_a.$$

## 4. COMPARISON WITH CLASSICAL SEPARATION AXIOMS

Recall that if a topological space  $S$  is  $T_1$  and strongly stacked, then

$$w_G = v_G.$$

where  $G \in \mathcal{OS}^\wedge$ . Moreover, every  $T_1$  stacked space is sober. This naturally raises the question of whether analogous results hold in the point-free setting. In this section, we investigate this question.

**Proposition 4.1.** *Let  $A$  be a frame. If*

$$v_F(0) = w_F(0)$$

*for every  $F \in A^\wedge$ , then every point of  $A$  is maximal in  $\text{pt } A$ .*

*Proof.* Let  $p, q \in \text{pt } A$  be such that  $p \leq q$ , and consider the open filter

$$F_q = \{x \in A \mid x \not\leq q\}.$$

Since  $A \setminus F_q = \downarrow q$ , it follows directly from the definition of  $w_{F_q}$  that

$$(3) \quad w_{F_q} = w_q.$$

In particular,

$$w_{F_q}(0) = q.$$

By hypothesis,

$$q = w_{F_q}(0) = v_{F_q}(0) = \bigvee_{x \in F_q} \neg x.$$

For every  $x \in F_q$ , we have  $x \not\leq q$ . Since  $p \leq q$ , it follows that  $x \not\leq p$ . Moreover,

$$x \wedge \neg x = 0 \leq p.$$

Since  $p$  is a point, it is meet-irreducible, and therefore

$$\neg x \leq p.$$

Thus,

$$q = v_{F_q}(0) = \bigvee_{x \in F_q} \neg x \leq p.$$

Since  $p \leq q$ , we conclude that  $p = q$ . Hence, every point of  $A$  is maximal in  $\text{pt } A$ .  $\square$

**Proposition 4.2.** *Let  $A$  be a frame. The following statements hold:*

(1) *If  $A$  is  $T_1$  and stacked, then*

$$v_F(0) = w_F(0)$$

*for every  $F \in A^\wedge$ .*

(2) *If*

$$v_F(0) = w_F(0)$$

*for every  $F \in A^\wedge$ , then  $A$  is stacked.*

(3) *Suppose that*

$$v_F(0) = w_F(0)$$

*for every  $F \in A^\wedge$  and that*

$$\{q \in \text{pt } A \mid a \leq q\} \neq \emptyset$$

*for every  $a \in A$ . Then  $A$  is  $T_1$ .*

*Proof.*

(1) Let  $(F, Q)$  be an HM pair. Since  $A$  is  $T_1$ , the space  $S = \text{pt } A$  is  $T_1$ . Hence  $Q = M$ . Since  $A$  is stacked,

$$\begin{aligned} v_F(0) &= (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) \\ &= (\mathcal{U}_* \circ [M'] \circ \mathcal{U})(0) \\ &= w_F(0). \end{aligned}$$

(2) Let  $(F, Q)$  be an HM pair. Recall that

$$(4) \quad v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F.$$

Evaluating at 0, we obtain

$$v_F(0) \leq (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) \leq w_F(0).$$

By hypothesis,  $v_F(0) = w_F(0)$ , and hence

$$v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0).$$

Therefore,  $A$  is stacked.

(3) Let  $p \in \text{pt } A$  and  $a \in A$  be such that

$$p \leq a.$$

By hypothesis, there exists  $q \in \text{pt } A$  such that

$$a \leq q.$$

Hence,

$$p \leq a \leq q.$$

By Proposition 4.1, every point of  $A$  is maximal. Therefore,  $p = q$ , and consequently

$$p \leq a \leq p.$$

Thus,  $a = p$ , and hence  $A$  is  $T_1$ . □

Thus, in the  $T_1$  case, the stacked condition forces the lower and upper bounds of every admissibility interval to agree at 0. The strongly stacked case yields the corresponding equality of the nuclei themselves.

**Corollary 4.3.** *Let  $A$  be a frame. The following statements hold:*

(1) *If  $A$  is  $T_1$  and strongly stacked, then*

$$v_F = w_F$$

*for every  $F \in A^\wedge$ .*

(2) *If*

$$v_F = w_F$$

*for every  $F \in A^\wedge$ , then  $A$  is strongly stacked.*

(3) *Suppose that*

$$v_F = w_F$$

*for every  $F \in A^\wedge$  and that*

$$\{q \in \text{pt } A \mid a \leq q\} \neq \emptyset$$

*for every  $a \in A$ . Then  $A$  is  $T_1$ .*

*Proof.*

(1) Let  $(F, Q)$  be an HM pair. Since  $A$  is strongly stacked,

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Moreover, since  $A$  is  $T_1$ , we have  $Q = M$ . Hence, by Proposition 3.2,

$$v_F = \mathcal{U}_* \circ [M'] \circ \mathcal{U} = w_F.$$

(2) By (4),

$$v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F.$$

Thus, if  $v_F = w_F$ , then

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U},$$

and therefore  $A$  is strongly stacked.

(3) Since  $v_F = w_F$  implies

$$v_F(0) = w_F(0),$$

the result follows from Proposition 4.2 (3). □

While the strongly stacked condition yields the collapse of every admissibility interval, the stacked condition already guarantees this phenomenon for those associated with completely prime filters. This gives the following consequence.

**Corollary 4.4.** *Let  $A$  be a  $T_1$  stacked frame. Then every admissibility interval associated with a completely prime filter collapses to a singleton. More precisely, for every  $p \in \text{pt } A$ ,*

$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

*Proof.* Let  $p \in \text{pt } A$ . Since  $A$  is  $T_1$  and stacked, Proposition 4.2 gives

$$v_{F_p}(0) = w_{F_p}(0) = p.$$

Moreover, as  $p$  is maximal,

$$w_{F_p} = w_p = u_p.$$

Recall that, for every nucleus  $j$ , one has

$$u_{j(0)} \leq j.$$

Indeed, for every  $x \in A$ ,

$$j(0) \vee x \leq j(x).$$

Applying this to  $j = v_{F_p}$  yields

$$u_p = u_{v_{F_p}(0)} \leq v_{F_p}.$$

On the other hand, by definition of the admissibility interval,

$$v_{F_p} \leq w_{F_p} = u_p.$$

Therefore,

$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

□

Notice that the additional hypothesis in Proposition 4.2 (3) is automatically satisfied whenever  $A$  is spatial. Indeed, if  $a < 1$  and  $A$  is spatial, then there exists  $p \in \text{pt } A$  such that

$$a \leq p.$$

Hence,

$$\mathcal{V}(a) = \{p \in \text{pt } A \mid a \leq p\} \neq \emptyset.$$

Thus, in the spatial setting, Proposition 6.5 reflects the topological fact that every  $T_1$  stacked space is sober.

The point-free situation is, however, genuinely broader, since the  $T_1$  and stacked conditions do not force spatiality. In fact, the following example shows that this remains true even if stacked is replaced by strongly stacked.

**Example 4.5.** Let  $B$  be an atomless complete Boolean algebra and consider

$$A = B \times 2,$$

where  $2 = \{0 < 1\}$ . Then  $A$  is a complete Boolean algebra. We show that  $A$  is  $T_1$  and strongly stacked, but not spatial.

First, let

$$p = (1, 0).$$

We claim that

$$\text{pt } A = \{p\}.$$

Indeed, suppose that  $(a, b), (c, d) \in A$  satisfy

$$(a, b) \wedge (c, d) \leq p.$$

Then

$$(a \wedge c, b \wedge d) \leq (1, 0),$$

so  $b \wedge d = 0$ . Since  $b, d \in 2$ , either  $b = 0$  or  $d = 0$ , and hence

$$(a, b) \leq p \quad \text{or} \quad (c, d) \leq p.$$

Thus,  $p \in \text{pt } A$ .

Now, let  $q = (x, y) \in \text{pt } A$ . If  $y = 0$ , then

$$(1, 0) \wedge (x, 1) \leq (x, 0) = q.$$

Since  $(x, 1) \not\leq q$ , the complete meet-irreducibility of  $q$  implies

$$(1, 0) \leq q,$$

and therefore  $x = 1$ . Hence  $q = p$ .

If  $y = 1$ , then  $x < 1$ . For  $a, b \in B$  such that  $a \wedge b \leq x$ , we have

$$(a, 1) \wedge (b, 1) \leq (x, 1) = q.$$

Since  $q$  is a point, either  $a \leq x$  or  $b \leq x$ , showing that  $x \in \text{pt } B$ . This is impossible, since an atomless complete Boolean algebra has no points. Therefore,

$$\text{pt } A = \{p\}.$$

Moreover, if

$$p \leq (a, b),$$

then  $a = 1$ , and hence  $(a, b)$  is either  $p$  or  $1$ . Thus  $p$  is maximal and  $A$  is  $T_1$ .

On the other hand,  $A$  is not spatial. Indeed, if  $a, b \in B$  with  $a \neq b$ , then

$$(a, 0) \neq (b, 0),$$

while

$$\mathcal{U}(a, 0) = \emptyset = \mathcal{U}(b, 0).$$

Hence the spatial reflection

$$\mathcal{U}: A \longrightarrow \mathcal{O}(\text{pt } A)$$

is not injective.

It remains to prove that  $A$  is strongly stacked. Since  $B$  is an atomless complete Boolean algebra,  $0$  is its only compact element. Therefore,

$$K(A) = K(B) \times K(2) = \{(0, 0), (0, 1)\}.$$

Since every open filter of a complete Boolean algebra is principal and  $\uparrow a$  is open if and only if  $a$  is compact, it follows that

$$A^\wedge = \{\uparrow(0, 0), \uparrow(0, 1)\}.$$

Clearly,

$$\uparrow(0, 0) = A,$$

whereas

$$\uparrow(0, 1) = B \times \{1\} = F_p.$$

Hence,

$$A^\wedge = \{A, F_p\}.$$

For  $F_p$ , since  $A$  is  $T_1$ , then

$$v_{F_p} = w_{F_p}.$$

For  $F = A$ , we have

$$v_F = \text{tp} = w_F.$$

Consequently,

$$v_F = w_F$$

for every  $F \in A^\wedge$ . By Corollary 4.3,  $A$  is strongly stacked.

**Corollary 4.6.** *There exist non-spatial  $T_1$  strongly stacked frames. In particular, there exist non-spatial  $T_1$  stacked frames.*

The previous example shows that the  $T_1$  and stacked conditions do not imply spatiality. We now compare stackedness with the stronger separation property **(H)**. Recall that a frame  $A$  satisfies property **(H)** if

$$1 \neq a \not\leq b$$

implies that there exists  $c \in A$  such that

$$c \not\leq a \quad \text{and} \quad \neg c \not\leq b.$$

As shown in [4], property **(H)** implies  $T_1$ . Nevertheless, property **(H)** does not imply stackedness, as the following example shows.

**Example 4.7.** There exists a frame satisfying property **(H)** which is not stacked.

Let  $I = [0, 1]$ , endowed with its usual topology, and consider the frame

$$A = \mathcal{O}(I).$$

Since  $I$  is a compact Hausdorff space,  $A$  is a compact Hausdorff frame. Moreover, since  $I$  is  $T_1$ , every point of  $A$  is maximal and

$$D(A) = \text{pt } A = \{I \setminus \{x\} \mid x \in I\},$$

where  $D(A)$  denotes the set of maximal elements of  $A$ .

Following the construction of Paseka and Šmarda [4, Proposition 2.3], let

$$\tilde{A} = \{(u, v) \in A \times A_{\neg\neg} \mid u \leq v\},$$

where

$$A_{\neg\neg} = \{a \in A \mid w_0(a) = a\}$$

is the Boolean algebra of regular elements of  $A$ . By [4, Propositions 2.3 and 2.4],  $\tilde{A}$  is a compact frame. In particular,  $\tilde{A}$  satisfies property **(H)**.

The maximal elements of  $\tilde{A}$  are precisely the elements of the form

$$D(\tilde{A}) = \{(d, 1) \mid d \in D(A)\}.$$

Since  $\tilde{A}$  satisfies **(H)**, it is  $T_1$ , and hence its points are precisely its maximal elements. Therefore,

$$\text{pt } \tilde{A} = \{(I \setminus \{x\}, I) : x \in I\}.$$

Since  $\tilde{A}$  is compact,

$$F = \uparrow 1 = \{1\}$$

is an open filter. Let  $(F, Q)$  be the corresponding HM pair. By the Hofmann–Mislove correspondence,

$$Q = \text{pt } \tilde{A} - F.$$

Since  $1 \notin \text{pt } \tilde{A}$ , it follows that

$$Q = \text{pt } \tilde{A}.$$

Now, since  $F = \{1\}$ ,

$$f_F = v_1 = \text{id}_{\tilde{A}},$$

and therefore

$$v_F(0) = 0_{\tilde{A}}.$$

On the other hand,

$$\begin{aligned} (U_* \circ [Q'] \circ U)(0) &= \bigwedge Q \\ &= \bigwedge \text{pt } \tilde{A} \\ &= (\emptyset, I) \neq (\emptyset, \emptyset) = v_F(0). \end{aligned}$$

Thus,

$$v_F(0) \neq (U_* \circ [Q'] \circ U)(0),$$

and consequently  $\tilde{A}$  is not stacked.

Consequently, although property **(H)** implies  $T_1$ , it does not imply stackedness. Indeed, the preceding example provides a compact Hausdorff frame which is not stacked.

Together with the strongly stacked non-spatial example above, this further illustrates the distinction between separation properties and stackedness in the point-free setting. On the one hand, there exist non-spatial  $T_1$  strongly stacked frames; on the other hand, even the stronger Hausdorff condition **(H)** does not guarantee stackedness.

## 5. QUOTIENTS AND PERMANENCE PROPERTIES

Simmons shows in [12, Lemma 8.9 and Corollary 8.10] that the following diagram

$$(5) \quad \begin{array}{ccc} A & \xrightarrow{f^\infty} & A \\ U_A \downarrow & & \downarrow U_A \\ \mathcal{O}S & \xrightarrow{F^\infty} & \mathcal{O}S \end{array}$$

commutes laxly, that is,  $U_A \circ f^\infty \leq F^\infty \circ U_A$ . In this diagram  $f^\infty$  and  $F^\infty$  represent the fitted nuclei associated with the filters  $F \in A^\wedge$  and  $\nabla \in \mathcal{O}S^\wedge$ , respectively, that is,

$$f^\infty = v_F \quad \text{and} \quad F^\infty = v_\nabla.$$

Here we prove something more general, since we consider  $j \in NA$ ,  $F \in A_j^\wedge$  and  $j_*F \in A^\wedge$  to produce the diagram

$$\begin{array}{ccc} A & \xrightarrow{f^\infty} & A \\ j^* \downarrow & & \downarrow j^* \\ A_j & \xrightarrow{\hat{f}^\infty} & A_j \end{array}$$

where  $\hat{f}^\infty$  and  $f^\infty$  are the fitted nuclei associated with the filters  $j_*F$  and  $F$ , respectively.

The following lemma is part of the analysis developed in this work and constitutes a key tool in the construction of the commutative diagrams that follow.

**Lemma 5.1.** *For  $j = j^*$ ,  $f$  and  $\hat{f}$  as before, it holds that  $j \circ \hat{f} \leq f \circ j$ .*

*Proof.* Recall that, by (1)

$$\hat{f} = \bigvee \{v_y \mid y \in j_*F\} \quad \text{y} \quad f = \bigvee \{v_{j(y)} \mid j(y) \in F\}.$$

Let  $a \in A$  be. For every  $y \in j_*F$ , we have

$$j(v_y(a)) = j(y \succ a) \leq (j(y) \succ j(a)) = v_{j(y)}(j(a)) \leq f(j(a)),$$

where the last inequality follows from  $j(y) \in F$ . Therefore, since  $j$  preserves arbitrary joins,

$$j(\hat{f}(a)) = j\left(\bigvee \{v_y(a) \mid y \in j_*F\}\right) = \bigvee \{j(v_y(a)) \mid y \in j_*F\} \leq f(j(a)).$$

Hence,  $j \circ \hat{f} \leq f \circ j$ . □

We now prove the above for an arbitrary ordinal.

**Corollary 5.2.** *For  $j$ ,  $f$  and  $\hat{f}$  as before, it holds that  $j \circ \hat{f}^\alpha \leq f^\alpha \circ j$  where  $\alpha \in \text{Ord}$ .*

*Proof.* We proceed by transfinite induction on  $\alpha$ . If  $\alpha = 0$ , the result is immediate.

For the induction step, suppose that the inequality holds for  $\alpha$ . Then

$$j \circ \hat{f}^{\alpha+1} = j \circ \hat{f} \circ \hat{f}^\alpha \leq f \circ j \circ \hat{f}^\alpha \leq f \circ f^\alpha \circ j = f^{\alpha+1} \circ j,$$

where the first inequality follows from Lemma 5.1 and the second from the induction hypothesis.

Finally, let  $\lambda$  be a limit ordinal. For every  $a \in A$ ,

$$j(\hat{f}(a)) = j\left(\bigvee_{\alpha < \lambda} \hat{f}^\alpha\right) = \bigvee_{\alpha < \lambda} j(\hat{f}^\alpha(a)) \leq \bigvee_{\alpha < \lambda} f^\alpha(j(a)) = f^\lambda(j(a)).$$

where the inequality follows from the induction hypothesis. Hence

$$j \circ \hat{f}^\lambda \leq f^\lambda \circ j.$$

This completes the proof.  $\square$

**Proposition 5.3.** *If  $A$  is a neat frame, then  $A_j$  is a neat frame.*

*Proof.* Let  $A$  be  $\alpha$ -neat and let  $F \in A_j^\wedge$ . By Proposition 2.1,  $j_*F \in A^\wedge$ . Let

$$\hat{f} = \bigvee_{y \in j_*F} v_y \quad \text{and} \quad f = \bigvee_{x \in F} v_x.$$

be the prenuclei associated with  $j_*F$  and  $F$ , respectively.

Since  $A$  is neat, there exists an ordinal  $\alpha$  such that, for every  $y \in j_*F$ ,

$$y \vee \hat{d} = 1, \quad \text{with } \hat{d} = \hat{f}^\alpha(0).$$

Let

$$d = f^\alpha(j(0)),$$

by Corollary 5.2,

$$j(\hat{d}) = j(\hat{f}^\alpha(0)) \leq f^\alpha(j(0)) = d.$$

Since  $j$  is inflationary,  $\hat{d} \leq j(\hat{d})$ , and hence

$$\hat{d} \leq d.$$

Now, let  $x \in F$ . Thus,  $x \in A_j$ , we have  $j(x) = x$ , and therefore  $x \in j_*F$ . Then,

$$x \vee \hat{d} = 1.$$

As  $\hat{d} \leq d$ , it follows that

$$1 = x \vee \hat{d} \leq x \vee d \leq 1.$$

Hence  $x \vee d = 1$ . Therefore,  $A_j$  is  $\alpha$ -neat.  $\square$

The following theorem establishes that tidiness is preserved under binary coproducts.

**Proposition 5.4.** *Let  $A_1$  and  $A_2$  be  $\alpha$ -neat frames. Then  $A_1 \oplus A_2$  is  $\alpha$ -neat.*

*Proof.* By construction of the coproduct  $A_1 \oplus A_2$ , consider the canonical frame morphism

$$\iota_i: A_i \rightarrow A_1 \oplus A_2, \quad i = 1, 2$$

together with their right adjoints  $(\iota_i)_*: A_1 \oplus A_2 \rightarrow A_i$ .

Then, we have the following diagram

$$\begin{array}{ccccccc} & & & \xrightarrow{\quad \iota_i \quad} & & & \\ & \nearrow & & & \searrow & & \\ A_i & \xrightarrow{\alpha_i} & A_1 \times A_2 & \xrightarrow{\downarrow(-)} & \mathcal{L}(A_1 \times A_2) & \xrightarrow{\sim} & A_1 \oplus A_2 \\ & \nwarrow & & & \nearrow & & \\ & & & \xleftarrow{\quad (\iota_i)_* \quad} & & & \end{array}$$

By Proposition 2.1,

$$(\iota_i)_* F \in A_i^\wedge.$$

Since  $A_i$  is  $\alpha$ -neat, for every  $x_i \in (\iota_i)_* F$  there exists  $d_i \in A_i$  such that

$$x_i \vee d_i = 1,$$

where

$$d_i = f_i^\alpha(0) \quad \text{and} \quad f_i = \bigvee_{y \in (\iota_i)_* F} v_y.$$

By the definition of  $(\iota_i)_* F$ ,

$$x_i \in (\iota_i)_* F \iff \iota_i(x_i) \in F.$$

Thus,

$$x_1 \oplus 1, 1 \oplus x_2 \in F.$$

Since  $F$  is a filter,

$$(x_1 \oplus 1) \wedge (1 \oplus x_2) = x_1 \oplus x_2 \in F.$$

Now let

$$f_F = \bigvee_{z \in F} v_z$$

be the prenucleus associated with  $F$ . We first show that, for  $i = 1, 2$ ,

$$\iota_i \circ f_i \leq f_F \circ \iota_i.$$

Indeed, let  $a \in A_i$  and  $y \in (\iota_i)_* F$ . Since  $\iota_i(y) \in F$ ,

$$\begin{aligned} \iota_i(v_y(a)) &= \iota_i(y \succ a) \\ &\leq (\iota_i(y) \succ \iota_i(a)) \\ &= v_{\iota_i(y)}(\iota_i(a)) \\ &\leq f_F(\iota_i(a)). \end{aligned}$$

Therefore, since the join defining  $f_i$  is pointwise and  $\iota_i$  preserves arbitrary joins,

$$\iota_i(f_i(a)) = \iota_i\left(\bigvee_{y \in (\iota_i)_* F} v_y(a)\right) = \bigvee_{y \in (\iota_i)_* F} \iota_i(v_y(a)) \leq f_F(\iota_i(a)).$$

Hence,

$$\iota_i \circ f_i \leq f_F \circ \iota_i.$$

We now prove by transfinite induction that

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal  $\beta$ .

If  $\beta = 0$ , the result is immediate. Suppose that it holds for  $\beta$ . Then

$$\begin{aligned} \iota_i \circ f_i^{\beta+1} &= \iota_i \circ f_i \circ f_i^\beta \\ &\leq f_F \circ \iota_i \circ f_i^\beta \\ &\leq f_F \circ f_F^\beta \circ \iota_i \\ &= f_F^{\beta+1} \circ \iota_i. \end{aligned}$$

Now let  $\lambda$  be a limit ordinal. For every  $a \in A_i$ ,

$$\begin{aligned} \iota_i(f_i^\lambda(a)) &= \iota_i\left(\bigvee_{\beta < \lambda} f_i^\beta(a)\right) \\ &= \bigvee_{\beta < \lambda} \iota_i(f_i^\beta(a)) \\ &\leq \bigvee_{\beta < \lambda} f_F^\beta(\iota_i(a)) \\ &= f_F^\lambda(\iota_i(a)). \end{aligned}$$

Thus,

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal  $\beta$ .

Evaluating at  $0 \in A_i$  for  $\beta = \alpha$ , we obtain

$$\iota_1(d_1) \leq f_F^\alpha(0 \oplus 1) \quad \text{and} \quad \iota_2(d_2) \leq f_F^\alpha(1 \oplus 0).$$

Therefore,

$$\begin{aligned} d_1 \oplus d_2 &= (d_1 \oplus 1) \wedge (1 \oplus d_2) \\ &\leq f_F^\alpha(0 \oplus 1) \wedge f_F^\alpha(1 \oplus 0) \\ &= f_F^\alpha(0 \oplus 1 \wedge 1 \oplus 0). \end{aligned}$$

Set  $d = f_F^\alpha(0 \oplus 0)$ , then  $d_1 \oplus d_2 \leq d$ . Finally,

$$\begin{aligned} 1 \oplus 1 &= (x_1 \vee d_1) \oplus (x_2 \vee d_2) \\ &= (x_1 \oplus x_2) \vee (d_1 \oplus d_2) \\ &\leq (x_1 \oplus x_2) \vee d \\ &\leq 1 \oplus 1. \end{aligned}$$

Therefore,

$$(x_q \oplus x_2) \vee d = 1 \oplus 1$$

and  $A_1 \oplus A_2$  is  $\alpha$ -neat. □

**Corollary 5.5.** *Let  $A_1$  and  $A_2$  be  $\alpha$ -neat and  $\beta$ -neat, respectively. Then  $A_1 \oplus A_2$  is  $\gamma$ -neat, where*

$$\gamma = \max\{\alpha, \beta\}.$$

*Proof.* Let  $\gamma = \max\{\alpha, \beta\}$ . Then  $A_1$  and  $A_2$  are both  $\gamma$ -neat. Hence, by Proposition 5.4,  $A_1 \oplus A_2$  is  $\gamma$ -neat.  $\square$

We extend the previous result to arbitrary coproducts.

**Proposition 5.6.** *Tidiness is preserved under arbitrary coproducts.*

*Proof.* Let  $\{A_i\}_{i \in I}$  be a family of neat frames, and for each  $i \in I$ , let  $\alpha_i$  be a tidiness degree of  $A_i$ . Consider

$$A = \bigoplus_{i \in I} A_i$$

and let

$$\iota_i: A_i \rightarrow A$$

denote the canonical coproduct morphisms. Since  $\iota_i$  is a frame morphism, it has a right adjoint  $(\iota_i)_*$ .

Let  $F \in A^\wedge$ . By Proposition 2.1,

$$(\iota_i)_* F \in A_i^\wedge$$

for every  $i \in I$ . Moreover,

$$x_i \in (\iota_i)_* F \iff \iota_i(x_i) \in F.$$

Set  $\alpha = \sup\{\alpha_i \mid i \in I\}$ . Then every  $A_i$  is  $\alpha$ -neat. Hence, for every  $x_i \in (\iota_i)_* F$ ,

$$x_i \vee d_i = 1,$$

where

$$d_i = f_i^\alpha(0) \quad \text{and} \quad f_i = \bigvee_{y \in (\iota_i)_* F} v_y.$$

Let

$$f_F = \bigvee_x F v_x$$

be the prenucleus associated with  $F$ . As in the proof of Proposition 5.4, for every  $i \in I$  we have

$$\iota_i \circ f_i \leq f_F \circ \iota_i,$$

and, by transfinite induction,

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal  $\beta$ .

Evaluating at  $0 \in A_i$  for  $\beta = \alpha$ , we obtain

$$\iota_i(d_i) = \iota_i(f_i^\alpha(0)) \leq f_F^\alpha(\iota_i(0)) = f_F^\alpha(\langle 0 \rangle).$$

Set

$$d = f_F^\alpha(\langle 0 \rangle).$$

Therefore,

$$\iota_i(d_i) \leq d$$

for every  $i \in I$ . Taking arbitrary joins yields

$$\langle d_i \rangle \bigvee_{i \in I} \iota_i(d_i) \leq d.$$

Now let

$$\langle x_i \rangle \in F.$$

Since  $x_i \vee d_i = 1$  for every  $i \in I$ , we have

$$\langle x_i \rangle \vee \langle d_i \rangle = \langle x_i \vee d_i \rangle = \langle 1 \rangle.$$

As  $\langle d_i \rangle \leq d$ ,

$$\langle 1 \rangle = \langle x_i \rangle \vee \langle d_i \rangle \leq \langle x_i \rangle \vee d \leq \langle 1 \rangle.$$

Thus,

$$\langle x_i \rangle \vee d = \langle 1 \rangle.$$

Therefore,  $A = \bigoplus_{i \in I} A_i$  is  $\alpha$ -neat, and hence neat.  $\square$

**Corollary 5.7.** *The subcategory of neat frames is a coreflective subcategory of  $\mathbf{Frm}$ .*

Recalling that  $v_F = f^\infty$  and  $v_{j_*F} = \hat{f}^\infty$ , Corollary 5.2 yields

$$j \circ v_{j_*F} \leq v_F \circ j$$

We now compare the compact quotients induced by  $F$  and  $j_*F$ , and construct a frame morphism between them making the corresponding diagram commute.

If we consider  $j, v_F$  and  $v_{j_*F}$  we have the following diagram

$$(6) \quad \begin{array}{ccc} A & \begin{array}{c} \xrightarrow{(v_{j_*F})^*} \\ \xleftarrow{(v_{j_*F})_*} \end{array} & A_{j_*F} \\ & \searrow H & \vdots h \\ j \downarrow & & \downarrow \\ A_j & \begin{array}{c} \xleftarrow{(v_F)_*} \\ \xrightarrow{(v_F)^*} \end{array} & A_F \end{array}$$

where  $A_F$  and  $A_{j_*F}$  are the compact quotients produced by  $v_F$  and  $v_{j_*F}$ , respectively.

Define

$$H: A \rightarrow A_F, \quad H = v_F \circ j.$$

Since  $v_F$  is idempotent,  $H(A) \subseteq A_F$ . Hence, by restricting  $H$  to  $A_{j_*F}$ , we obtain a map

$$h: A_{j_*F} \rightarrow A_F, \quad h = H|_{A_{j_*F}}.$$

As finite meets in quotient frames are inherited from the ambient frame,  $h$  preserves finite meets. It remains to verify that  $h$  preserves arbitrary joins.

**Lemma 5.8.** *The map  $h: A_{j_*F} \rightarrow A_F$  preserves arbitrary joins.*

*Proof.* Let  $X \subseteq A_{j_*F}$ . Since joins in  $A_{j_*F}$  are given by

$$\bigvee_{v_{j_*F}} X = v_{j_*F}(\bigvee X)$$

we have

$$\begin{aligned} h\left(\bigvee_{v_{j_*F}} X\right) &= v_F\left(j\left(v_{j_*F}\left(\bigvee X\right)\right)\right) \\ &\leq v_F\left(v_F\left(j\left(\bigvee X\right)\right)\right) \\ &= v_F\left(j\left(\bigvee X\right)\right), \end{aligned}$$

where the inequality follows from Corollary 5.2 and the equality from the idempotency of  $v_F$ .

Since  $H$  is a frame morphism,

$$v_F\left(j\left(\bigvee X\right)\right) = \bigvee_{v_F}\{H(x) \mid x \in X\} = \bigvee_{v_F}\{h(x) \mid x \in X\}.$$

Therefore,

$$h\left(\bigvee_{v_{j_*F}} X\right) \leq \bigvee_{v_F} h[X].$$

Conversely, for every  $x \in X$ ,

$$x \leq \bigvee_{v_{j_*F}} X,$$

and hence, by monotonicity of  $h$ ,

$$h(x) \leq h\left(\bigvee_{v_{j_*F}} X\right).$$

Since this holds for every  $x \in X$ , we have

$$\bigvee_{v_F} h[X] \leq h\left(\bigvee_{v_{j_*F}} X\right).$$

Thus,

$$h\left(\bigvee_{v_{j_*F}} X\right) = \bigvee_{v_F} h[X].$$

Therefore  $h$  preserves arbitrary joins.  $\square$

The previous construction yields the following commutative diagram induced by the compact quotients associated with the fitted nuclei.

**Proposition 5.9.** *The diagram*

$$(7) \quad \begin{array}{ccc} A & \xrightarrow{v_{j_*F}} & A_{j_*F} \\ j \downarrow & & \downarrow h \\ A_j & \xrightarrow{v_F} & A_F \end{array}$$

is commutative.

*Proof.* By Lemma 5.8,  $h$  is a frame morphism. The commutativity of the diagram follows from the definition of  $h$ .  $\square$

We now study the behavior of strongly stacked frames under quotients.

Recall that, for a frame morphism  $f: A \rightarrow B$ ,  $j \in NA$ , and  $k \in NB$ , we have

$$N(f)(j) = \bigvee \{u_{f(j(a))} \wedge v_{f(a)} \mid a \in A\} \in NB$$

and

$$N(f)_*(k) = f_* \circ k \circ f \in NA,$$

where  $f_*$  denotes the right adjoint of  $f$ .

**Proposition 5.10.** *Let  $A$  be a strongly stacked frame and  $j \in NA$ . Then  $A_j$  is strongly stacked.*

*Proof.* Let  $F \in A_j^\wedge$ , and set

$$S = \text{pt } A, \quad S_j = \text{pt } A_j.$$

Consider the open filter

$$G = j_*(F) = \{x \in A \mid j(x) \in F\},$$

and let

$$Q = S_j \setminus F, \quad R = S \setminus G.$$

Denote by

$$\mathcal{U}: A \longrightarrow \mathcal{OS} \quad \text{and} \quad \mathcal{U}_j: A_j \longrightarrow \mathcal{OS}_j$$

the corresponding spatial reflections.

Set

$$l = N(\mathcal{U}_j)_*([Q']) = (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j$$

and

$$m = N(j^*)(v_G).$$

By Proposition 2.4,

$$v_F \leq l.$$

Thus, it is enough to prove

$$l \leq m \leq v_F.$$

We first prove that  $m \leq v_F$ . Since  $G = j_*(F)$ , the fitted nuclei associated with  $G$  and  $F$  satisfy

$$(8) \quad j^* \circ v_G \leq v_F \circ j^*.$$

Moreover,

$$m = N(j^*)(v_G) = \bigvee_{x \in A} (u_{j^*(v_G(x))} \wedge v_{j^*(x)}).$$

By (8), for every  $x \in A$ ,

$$j^*(v_G(x)) \leq v_F(j^*(x)),$$

and hence

$$u_{j^*(v_G(x))} \leq u_{v_F(j^*(x))}.$$

Therefore,

$$u_{j^*(v_G(x))} \wedge v_{j^*(x)} \leq u_{v_F(j^*(x))} \wedge v_{j^*(x)} \leq v_F.$$

Taking joins over  $x \in A$ , we obtain

$$(9) \quad m \leq v_F.$$

We now prove that  $l \leq m$ . Since  $A$  is strongly stacked,

$$v_G = N(\mathcal{U})_*([R']) = \mathcal{U}_* \circ [R'] \circ \mathcal{U}.$$

Hence,

$$m = N(j^*)(N(\mathcal{U})_*([R'])).$$

To compare the  $u$ -components in the decompositions of  $l$  and  $m$ , consider the auxiliary nucleus

$$n = N(\mathcal{U}_j \circ j^*)_*([Q']).$$

We first observe that, for every  $x \in A_j$ ,

$$j^*(n(x)) = l(x).$$

Indeed,

$$\begin{aligned} j^*(n(x)) &= j^*(j_* \circ (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j \circ j^*)(x) \\ &= j^*j_* \left( \bigvee \{a \in A_j \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j^*(x))\} \right) \\ &= \bigvee \{a \in A_j \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(x)\} \\ &= l(x). \end{aligned}$$

Now, under the usual identification of  $S_j$  with the corresponding subspace of  $S$ , we have

$$Q = R \cap S_j.$$

Using the decomposition of a nucleus in terms of its  $u$ - and  $v$ -components, we have

$$l = \bigvee_{x \in A_j} (u_{l(x)} \wedge v_x) = \bigvee_{x \in A_j} (u_{j^*(n(x))} \wedge v_x).$$

On the other hand, since

$$m = N(j^*)(N(\mathcal{U})_*([R'])),$$

we have

$$m = \bigvee_{a \in A} (u_{j^*(N(\mathcal{U})_*([R'])(a))} \wedge v_{j^*(a)}).$$

For  $x \in A_j$ , the explicit description of the right adjoint gives

$$\begin{aligned} j^*(N(\mathcal{U})_*([R'])(x)) &= j^* \left( \bigvee \{a \in A \mid \mathcal{U}(a) \subseteq R' \cup \mathcal{U}(x)\} \right) \\ &= \bigvee \{j^*(a) \in A_j \mid \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\}. \end{aligned}$$

Moreover, since  $x \in A_j$ ,

$$\mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)$$

is equivalent, after restriction to  $S_j$ , to

$$\mathcal{U}_j(j^*(a)) \subseteq Q' \cup \mathcal{U}_j(x).$$

Consequently,

$$j^*(n(x)) \leq j^*(N(\mathcal{U})_*([R'])(x)).$$

Therefore,

$$u_{j^*(n(x))} \leq u_{j^*(N(\mathcal{U})_*([R'])(x))}$$

for every  $x \in A_j$ , and hence

$$u_{l(x)} \wedge v_x = u_{j^*(n(x))} \wedge v_x \leq u_{j^*(N(\mathcal{U})_*([R'])(x))} \wedge v_x \leq m.$$

Taking joins over  $x \in A_j$ , we obtain

$$l \leq m.$$

Combining this inequality with (9) and  $v_F \leq l$ , we obtain

$$v_F \leq l \leq m \leq v_F.$$

Hence,

$$v_F = l = (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j.$$

Therefore,  $A_j$  is strongly stacked.  $\square$

Since every strongly stacked frame is stacked, we obtain the following result.

**Corollary 5.11.** *Let  $A$  be a strongly stacked frame and  $j \in NA$ . Then  $A_j$  is stacked.*

*Proof.* By Proposition 5.10,  $A_j$  is strongly stacked, and hence stacked.  $\square$

We first observe that the KC property is inherited by quotients.

**Proposition 5.12.** *Let  $A$  be a KC frame and  $j \in NA$ . Then  $A_j$  is KC.*

*Proof.* Let  $k \in N(A_j)$  be such that  $(A_j)_k$  is compact. Then

$$\nabla(k) \in A_j^\wedge.$$

By Proposition 2.1,

$$j_*(\nabla(k)) \in A^\wedge.$$

Consider the nucleus

$$l = j_* \circ k \circ j^* \in NA.$$

Since

$$\nabla(l) = j_*(\nabla(k)),$$

the quotient  $A_l$  is compact. Since  $A$  is KC, there exists  $a \in A$  such that

$$l = u_a.$$

We claim that

$$k = u_{j^*(a)}.$$

Indeed, from  $l = u_a$  we have

$$a \vee x = l(x) = j_*(k(j^*(x)))$$

for every  $x \in A$ . Applying  $j^*$  to both sides yields

$$j^*(a) \vee j^*(x) = k(j^*(x)),$$

since  $j^* \circ j_* = \text{id}_{A_j}$ .

Now let  $x \in A_j$ . Since  $j^*: A \rightarrow A_j$  is surjective, we have  $j^*(x) = x$ , and therefore

$$\begin{aligned} u_{j^*(a)}(x) &= j^*(a) \vee x \\ &= j^*(a) \vee j^*(x) \\ &= k(j^*(x)) \\ &= k(x). \end{aligned}$$

Hence

$$k = u_{j^*(a)}.$$

Thus  $(A_j)_k$  is closed. Therefore,  $A_j$  is KC. □

**Proposition 5.13.** *Every KC frame is  $T_1$ .*

*Proof.* Let  $p \in \text{pt } A$ . The filter

$$F_p = \nabla(w_p) = \{x \in A \mid x \not\leq p\}$$

is completely prime and, in particular,  $F_p \in A^\wedge$ . Hence  $A_{w_p}$  is a compact quotient of  $A$ . Since  $A$  is KC, this quotient is closed. Therefore,

$$w_p = u_a$$

for some  $a \in A$ . Evaluating at 0, we obtain

$$a = u_a(0) = w_p(0) = p,$$

and hence

$$w_p = u_p.$$

Now let  $a \in A$  be such that

$$p < a \leq 1.$$

Since  $a \not\leq p$ , we have

$$w_p(a) = 1.$$

On the other hand,

$$u_p(a) = p \vee a = a.$$

Since  $w_p = u_p$ , it follows that

$$a = 1.$$

Thus  $p$  is maximal. Since every point of  $A$  is maximal,  $A$  is  $T_1$ . □

## 6. THE SPATIAL CASE

**Lemma 6.1.** *Let  $\alpha \in \text{Ord}$  and  $X \in CS$ . Then*

$$(10) \quad \partial_F^\alpha(X) = (f_F^\alpha(X'))'.$$

*Proof.* We proceed by transfinite induction on  $\alpha$ .

For  $\alpha = 0$ , both  $\partial_F^0$  and  $f_F^0$  are the identity. Hence,

$$\partial_F^0(X) = X = (X')' = (f_F^0(X'))'.$$

Suppose that the result holds for some ordinal  $\alpha$ . Then

$$\begin{aligned} \partial_F^{\alpha+1}(X) &= \partial_F(\partial_F^\alpha(X)) \\ &= (f_F((\partial_F^\alpha(X))'))' \\ &= (f_F(f_F^\alpha(X')))' \\ &= (f_F^{\alpha+1}(X'))'. \end{aligned}$$

Finally, let  $\lambda$  be a limit ordinal and suppose that the result holds for every  $\alpha < \lambda$ . By definition and De Morgan's law,

$$\begin{aligned} (\partial_F^\lambda(X))' &= \left( \bigcap_{\alpha < \lambda} \partial_F^\alpha(X) \right)' \\ &= \bigcup_{\alpha < \lambda} (\partial_F^\alpha(X))' \\ &= \bigcup_{\alpha < \lambda} f_F^\alpha(X') \\ &= f_F^\lambda(X'), \end{aligned}$$

where the third equality follows from the induction hypothesis. Therefore,

$$\partial_F^\lambda(X) = (f_F^\lambda(X'))'.$$

This completes the proof.  $\square$

The following is an immediate consequence of Lemma 6.1.

**Corollary 6.2.** *Let  $U \in OS$ . Then*

$$(11) \quad (\partial_F^\infty(U'))' = v_F(U).$$

A particular instance of diagram (7) is obtained by taking  $j^* = \mathcal{U}_A$ , where  $S = \text{pt } A$ . Thus, we obtain the following commutative diagram:

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ \mathcal{U}_A \downarrow & & \downarrow h \\ OS & \xrightarrow{v_\nabla} & OS_\nabla \end{array}$$

Recall that  $\text{pt } A_F = Q$ . Hence, the spatial reflection of  $A_F$  is given by

$$\mathcal{U}_{A_F}: A_F \rightarrow \mathcal{O} \text{pt } A_F = \mathcal{O}Q.$$

Similarly, since  $\text{pt}((\mathcal{O}S)_\nabla) = Q$ , we have the spatial reflection

$$\mathcal{U}_{\mathcal{O}S_\nabla}: \mathcal{O}S_\nabla \rightarrow \mathcal{O} \text{pt}(\mathcal{O}S)_\nabla = \mathcal{O}Q.$$

Consequently, the two induced morphisms from  $A$  to  $\mathcal{O}Q$  are both given by

$$x \mapsto \mathcal{U}_A(x) \cap Q.$$

Therefore, the preceding constructions fit into the following commutative diagram:

$$(12) \quad \begin{array}{ccccc} A & \xrightarrow{v_F} & A_F & & \\ \downarrow \mathcal{U}_A & & \downarrow h & \searrow \mathcal{U}_{A_F} & \\ \mathcal{O}S & \xrightarrow{v_\nabla} & \mathcal{O}S_\nabla & \nearrow \mathcal{U}_{\mathcal{O}S_\nabla} & \mathcal{O}Q \end{array}$$

Since the above diagram is determined by  $F \in A^\wedge$  and the associated compact saturated subset  $Q \in \mathcal{Q}S$ , we refer to diagram (12) as the  $Q$ -square.

**Theorem 6.3.** *Under the hypotheses of the  $Q$ -square, if  $A$  satisfies property **(H)**, then*

$$\mathcal{O}Q \cong \uparrow Q'.$$

*Equivalently, the frame of open sets of the point space of  $A_F$  is isomorphic to a compact closed quotient of a Hausdorff space.*

*Proof.* Suppose that  $A$  satisfies **(H)**. Since **(H)** is conservative,  $S = \text{pt } A$  is a  $T_2$  space. Hence, for every  $Q \in \mathcal{Q}S$ , the set  $Q$  is closed, and therefore  $Q' \in \mathcal{O}S$ .

Moreover, since every  $T_2$  space is strongly stacked, we have

$$v_\nabla = [Q'].$$

As  $Q' \in \mathcal{O}S$ , it follows that

$$[Q'] = u_{Q'}$$

Consequently,

$$(\mathcal{O}S)_{v_\nabla} = (\mathcal{O}S)_{u_{Q'}} \cong \uparrow Q'.$$

Now,  $u_{Q'}$  is a complemented nucleus in  $\mathcal{O}S$ , by [5, Proposition VI.3.3], implies that  $\mathcal{O}S_{v_\nabla}$  is spatial. Therefore,

$$(\mathcal{O}S)_{v_\nabla} \cong \mathcal{O} \text{pt}((\mathcal{O}S)_{v_\nabla}).$$

Since

$$\text{pt}((\mathcal{O}S)_{v_\nabla}) = Q,$$

we obtain

$$\mathcal{O}Q \cong (\mathcal{O}S)_{v_\nabla} \cong \uparrow Q'.$$

□

We now show that both notions are conservative; namely, for every topological space  $S$ ,

$$S \text{ is stacked} \iff \mathcal{O}S \text{ is stacked},$$

and

$$S \text{ is strongly stacked} \iff \mathcal{O}S \text{ is strongly stacked}.$$

**Proposition 6.4.** *The notions of strongly stacked and stacked are conservative.*

*Proof.* Let  $S \in \mathbf{Top}$  and set  $A = \mathcal{O}S$ .

First, suppose that  $S$  is strongly stacked. Then, for every HM pair  $(F, Q)$ ,

$$v_F = [Q'].$$

By Lemma 5.3,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = [Q'].$$

Therefore,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = v_F,$$

and hence  $\mathcal{O}S$  is strongly stacked.

Now, suppose that  $S$  is stacked. Then, for every HM pair  $(F, Q)$ ,

$$Q = \partial_F^\infty(S).$$

Hence,

$$Q' = (\partial_F^\infty(S))'.$$

By Corollary 5.2,

$$v_F(\emptyset) = (\partial_F^\infty(S))' = Q'.$$

In particular,  $Q'$  is open. By Lemma 5.3,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = [Q'].$$

Therefore,

$$\begin{aligned} (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(\emptyset) &= [Q'](\emptyset) \\ &= (Q')^\circ \\ &= Q' \\ &= v_F(\emptyset). \end{aligned}$$

Thus,  $\mathcal{O}S$  is stacked.

Since all the implications used above are equivalences, the converse statements follow by reversing the argument.  $\square$

Recall that  $A^\wedge$  is endowed with the topology generated by the basic open sets

$$D(x) = \{F \in A^\wedge \mid x \in F\}, \quad x \in A.$$

Moreover, let  $VA$  denote the Vietoris modification of  $A$  introduced in [12]. The points of  $VA$  are of the form

$$(F, a), \quad F \in A^\wedge, \quad a \in [v_F(0), w_F(0)].$$

Consider the continuous surjection

$$\pi: \text{pt}(VA) \longrightarrow A^\wedge, \quad (F, a) \longmapsto F.$$

It induces the frame morphism

$$\mathcal{O}(\pi) = \pi^{-1}: \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\text{pt}(VA)).$$

Recall also that the subbasic open sets of  $\text{pt}(VA)$  are determined by

$$(13) \quad (F, a) \in \Box x \iff x \in F, \quad (F, a) \in \Diamond x \iff x \not\leq a.$$

Particularly,

$$(14) \quad a \in [F](x) \iff x \in F \quad \text{and} \quad a \in \langle F \rangle(x) \iff x \not\leq a.$$

**Proposition 6.5.** *Let  $A$  be a frame, and consider the continuous surjection*

$$\pi: \text{pt}(VA) \longrightarrow A^\wedge, \quad (F, a) \longmapsto F.$$

*Then the following statements hold:*

- (1) *If  $A$  is  $T_1$  and stacked, then*

$$\mathcal{O}(\pi) = \pi^{-1}: \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\text{pt}(VA))$$

*is an isomorphism.*

- (2) *If  $\mathcal{O}(\pi)$  is an isomorphism, then  $A$  is stacked.*

- (3) *If  $\mathcal{O}(\pi)$  is an isomorphism and*

$$\mathcal{V}(a) = \{p \in \text{pt } A \mid a \leq p\} \neq \emptyset$$

*for every  $a < 1$ , then  $A$  is  $T_1$ .*

*Proof.*

- (1) Suppose that  $A$  is  $T_1$  and stacked. By Proposition 4.2,

$$v_F(0) = w_F(0)$$

for every  $F \in A^\wedge$ . Hence, each fiber of

$$\pi: \text{pt}(VA) \longrightarrow A^\wedge$$

contains a unique element. Therefore,  $\pi$  is bijective.

It remains to prove that

$$\pi^{-1}: A^\wedge \longrightarrow \text{pt}(VA)$$

is continuous. Recall that the subbasic open sets of  $\text{pt}(VA)$  are given by

$$(F, a) \in [-](x) \iff x \in F$$

and

$$(F, a) \in \langle - \rangle(x) \iff x \not\leq a.$$

For the first family,

$$(F, a) \in [-](x) \iff x \in F \iff F \in D(x),$$

and therefore

$$\pi([-](x)) = D(x).$$

Now consider  $\langle - \rangle(x)$ . Since the fibers of  $\pi$  collapse, for every  $(F, a) \in \text{pt}(VA)$  we have

$$a = v_F(0).$$

Thus,

$$\begin{aligned} (F, a) \notin \langle - \rangle(x) &\iff x \leq a \\ &\iff x \leq v_F(0) \\ &\iff v_F(x) = v_F(0) \\ &\iff x \notin F. \end{aligned}$$

Hence,

$$(F, a) \in \langle - \rangle(x) \iff x \in F \iff F \in D(x),$$

and therefore

$$\pi(\langle - \rangle(x)) = D(x).$$

Thus, the image under  $\pi$  of every subbasic open set of  $\text{pt}(VA)$  is open in  $A^\wedge$ . Hence  $\pi^{-1}$  is continuous, and consequently  $\pi$  is a homeomorphism. Therefore,

$$\mathcal{O}(\pi) = \pi^{-1}: \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\text{pt}(VA))$$

is an isomorphism.

(2) Suppose that

$$\mathcal{O}(\pi) = \pi^{-1}: \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\text{pt}(VA))$$

is an isomorphism. Since  $\pi^{-1}$  is bijective,  $\pi$  is injective. For every  $F \in A^\wedge$ ,

$$(F, v_F(0)), (F, w_F(0)) \in \text{pt}(VA)$$

and

$$\pi(F, v_F(0)) = F = \pi(F, w_F(0)).$$

Hence,

$$v_F(0) = w_F(0).$$

By Proposition 4.2,  $A$  is stacked.

(3) Suppose that  $\mathcal{O}(\pi)$  is an isomorphism and that

$$\mathcal{V}(a) = \{q \in \text{pt } A \mid a \leq q\} \neq \emptyset$$

for every  $a < 1$ . By part (2),

$$v_F(0) = w_F(0)$$

for every  $F \in A^\wedge$ . Therefore, by Proposition 4.1, every point of  $A$  is maximal.

Let  $p \in \text{pt } A$  and  $a \in A$  be such that

$$p \leq a.$$

By hypothesis, there exists  $q \in \text{pt } A$  such that

$$a \leq q.$$

Thus,

$$p \leq a \leq q.$$

Since  $p$  and  $q$  are maximal, we have  $p = q$ , and therefore

$$p \leq a \leq p.$$

Hence  $a = p$ , and  $A$  is  $T_1$ . □

## 7. RELATIONS AMONG COMPACT-QUOTIENT SEPARATION PROPERTIES

For every HM triple  $(F, Q, G)$ , we have

$$\mathcal{U} \circ v_F \leq v_G \circ \mathcal{U}.$$

Suppose now that  $A$  is strongly stacked. Then

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Since

$$v_G \leq [Q'],$$

we obtain

$$\mathcal{U}_* \circ v_G \circ \mathcal{U} \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} = v_F.$$

Together with the previous inequality, this yields

$$v_F = \mathcal{U}_* \circ v_G \circ \mathcal{U},$$

or equivalently,

$$\mathcal{U} \circ v_F = v_G \circ \mathcal{U}.$$

Thus, if  $A$  is strongly stacked, the corresponding diagram is commutative.

**Proposition 7.1.** *Let  $A$  be a neat frame. If*

$$\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$$

*for every HM triple  $(F, Q, \nabla)$ , then  $A$  is strongly stacked.*

*Proof.* Let  $S = \text{pt } A$  and let  $(F, Q, \nabla)$  be an HM triple.

Since  $A$  is neat,  $\mathcal{O}S$  is neat. Hence  $S$  is stacked and packed. In particular, every compact saturated subset  $Q \subseteq S$  is closed, so

$$Q' \in \mathcal{O}S.$$

Since  $S$  is stacked,

$$Q = \partial_\nabla^\infty(S),$$

and therefore, by Corollary 6.2,

$$v_\nabla(\emptyset) = Q'.$$

Because  $\mathcal{O}S$  is neat, it follows that

$$v_\nabla = u_{Q'} = [Q'].$$

By hypothesis,

$$\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U} = [Q'] \circ \mathcal{U}.$$

Applying the right adjoint  $\mathcal{U}_*$ , we obtain

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Therefore,  $A$  is strongly stacked. □

Consequently, we have the following result.

Ejemplo de neat  
no KC

Comprobar si  
neat implica  
stacked

KC implica S.S.?

Existe alguna  
propiedad P tal  
que (H)+P im-  
plica staked

T1+S.+P implica  
(H)?

Relacionar S. y  
S.S. con fit y sfit

Relacionar el  
trabajo con lo  
presentado por  
Guram y Sebas-  
tian

**Corollary 7.2.** *If  $A$  is a neat frame, then  $\text{pt } A$  is strongly stacked.*

*Proof.* Let  $S = \text{pt } A$ . Since  $A$  is neat,  $\mathcal{O}S$  is neat, and therefore  $S$  is strongly stacked.  $\square$

**Proposition 7.3.** *Let  $A$  be a KC-frame. Then  $A$  is neat.*

*Proof.* Let  $F \in A^\wedge$  and put

$$d = v_F(0).$$

Since  $A$  is KC and

$$v_F \in [v_F, w_F],$$

there exists  $a \in A$  such that

$$v_F = u_a.$$

Evaluating at 0, we obtain

$$d = v_F(0) = u_a(0) = a.$$

Hence

$$v_F = u_d.$$

Since this holds for every  $F \in A^\wedge$ , the frame  $A$  is neat.  $\square$

**Proposition 7.4.** *Let  $A$  be a KC-frame. Then  $A$  is stacked if and only if it is strongly stacked.*

*Proof.* Let  $(F, Q)$  be an HM-pair and put

$$d = v_F(0) \quad \text{and} \quad a = w_F(0).$$

Since every KC-frame is  $T_1$ , we have

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F.$$

Moreover, by the previous proposition,

$$v_F = u_d.$$

Since  $A$  is KC,  $w_F$  is closed as well, and therefore

$$w_F = u_a.$$

Consequently,

$$\begin{aligned} A \text{ is strongly stacked} &\iff v_F = w_F \\ &\iff u_d = u_a \\ &\iff d = a \\ &\iff v_F(0) = w_F(0) \\ &\iff A \text{ is stacked.} \end{aligned}$$

Since this holds for every HM-pair  $(F, Q)$ , the result follows.  $\square$

**Corollary 7.5.** *Let  $A$  be a KC-frame spatial. Then  $S$  is stacked if and only if it is strongly stacked.*

**Corollary 7.6.** *Let  $A$  be a spatial frame. Then  $A$  is KC if and only if it is neat.*

DONE: Agregar como corolario el caso espacial del resultado anterior.

*Proof.* Suppose first that  $A$  is KC. By the previous results, every KC-frame is neat. Conversely, let  $A$  be neat. Since  $A$  is spatial, we may write

$$A \cong \mathcal{O}(S)$$

for some topological space  $S$ . By the spatial characterization of neatness,  $S$  is packed and stacked. Moreover, neatness implies that  $S$  is strongly stacked. Hence  $S$  is packed and strongly stacked, and therefore  $\mathcal{O}(S)$  is KC. Thus  $A$  is KC.  $\square$

**Proposition 7.7.** *Let  $A$  be a KC-frame. If  $A$  is subfit, then  $A$  is strongly stacked.*

*Proof.* Let  $(F, Q)$  be an HM-pair and put

$$d = v_F(0) \quad \text{and} \quad a = w_F(0).$$

Since  $A$  is KC, both  $v_F$  and  $w_F$  are closed nuclei. Hence

$$v_F = u_d \quad \text{and} \quad w_F = u_a.$$

By subfitness and [14, Theorem 3.5], we have

$$d = a.$$

Therefore,

$$v_F = u_d = u_a = w_F.$$

Since  $A$  is  $T_1$ , we also have

$$w_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Consequently,

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Since this holds for every HM-pair  $(F, Q)$ ,  $A$  is strongly stacked.  $\square$

**Corollary 7.8.** *Every subfit KC-frame is spatial.*

*Proof.* Let  $A$  be a subfit KC-frame. Since every KC-frame is  $T_1$ , the result follows from [6, Proposition 2.5.3].  $\square$

**Proposition 7.9.** *There exists a KC-frame which is strongly stacked and not subfit. Consequently,*

$$\text{KC} \not\Rightarrow (\text{sfit}).$$

*Proof.* Consider the frame

$$A = B \times 2,$$

introduced above, where  $B$  is the complete Boolean algebra considered in the previous example. Recall that  $A$  is  $T_1$ , strongly stacked, and non-spatial.

The only open filters of  $A$  are

$$A \quad \text{and} \quad F_p = \uparrow(0, 1) = B \times \{1\},$$

where  $p = (1, 0)$  is the unique point of  $A$ .

For  $F = A$ , we have

$$v_A = w_A = \text{tp} = u_1.$$

Hence

$$[v_A, w_A] = \{u_1\}.$$

On the other hand, for  $F = F_p$ , the computations in the previous example give

$$v_{F_p} = w_{F_p} = u_p.$$

Therefore,

$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

Thus, for every  $F \in A^\wedge$ , every nucleus in the admissibility interval  $[v_F, w_F]$  is closed. Hence  $A$  is a KC-frame.

Finally,  $A$  cannot be subfit. Indeed, every  $T_1$  subfit frame is spatial, whereas  $A$  is  $T_1$  and non-spatial. Consequently,  $A$  is a KC-frame which is strongly stacked and not subfit. In particular,

$$\text{KC} \not\Rightarrow (\text{sfit}).$$

□

**Proposition 7.10.** *Every neat strongly stacked frame is KC.*

*Proof.* Let  $F \in A^\wedge$ . Since  $A$  is neat, it is  $T_1$ . Hence, as  $A$  is strongly stacked, we have

$$v_F = w_F.$$

Moreover, of tidiness,

$$v_F = u_d, \quad d = v_F(0).$$

Therefore,

$$[v_F, w_F] = \{u_d\}.$$

Since every nucleus inducing a compact quotient of  $A$  belongs to an admissibility interval  $[v_F, w_F]$  for some  $F \in A^\wedge$ , every such nucleus is closed. Hence every compact quotient of  $A$  is closed, and therefore  $A$  is KC. □

## REFERENCES

1. Francisco Ávila, Guram Bezhanishvili, PJ Morandi, and Angel Zaldívar, *The frame of nuclei on an alexandorff space*, Order (2020).
2. Melvin Hochster, *Prime ideal structure in commutative rings*, Transactions of the American Mathematical Society **142** (1969), 43–60.
3. P. T. Johnstone, *Stone spaces*, Cambridge Studies in Advanced Mathematics, vol. 3, Cambridge University Press, Cambridge, 1982. MR 698074
4. Jan Paseka and Bohumil Šmarda,  *$T_2$ -frames and almost compact frames*, Czechoslovak Mathematical Journal **42** (1992), no. 3, 385–402.
5. J. Picado and A. Pultr, *Frames and locales: Topology without points*, Frontiers in Mathematics, Springer Basel, 2012.
6. Jorge Picado and Aleš Pultr, *Separation in point-free topology*, Springer, 2021.
7. R. A. Sexton and H. Simmons, *An ordinal indexed hierarchy of separation properties*, Topology Proceedings **30** (2006), no. 2, 585–625.
8. RA Sexton and H Simmons, *Point-sensitive and point-free patch constructions*, Journal of Pure and Applied Algebra **207** (2006), no. 2, 433–468.
9. Rosemary A. Sexton, *The use of the frame theoretic assembly as a unifying construct*, Master's thesis, The Victoria University of Manchester, 2000.
10. ———, *A point-free and point-sensitive analysis of the patch assembly*, Ph.D. thesis, The Victoria University of Manchester, 2003.
11. Harold Simmons, *A collection of notes on frames*.

12. ———, *The vietoris modifications of a frame*, Unpublished manuscript, 79pp., available online at <http://www.cs.man.ac.uk/hsimmons> (2004).
13. ———, *The assembly of a frame*, University of Manchester (2006).
14. ———, *Regularity, fitness, and the block structure of frames*, Applied Categorical Structures **14** (2006), 1–34.

Following the idea presented in [14] (or, alternatively, in [12]), let us consider the linearly ordered frame

$$A = [0, 1].$$

In this case the following observations hold.

- (1) For all  $a, b \in A$  the implication  $(a \succ b)$  is given by

$$(a \succ b) = \begin{cases} 1, & \text{si } a \leq b, \\ b, & \text{si } a > b. \end{cases}$$

- (2) For  $a, x \in A$ , the distinguished nuclei  $u_a, v_a, w_a$  are given by

$$u_a(x) = \begin{cases} x & \text{si } a \leq x, \\ a & \text{si } x < a, \end{cases}, \quad v_a(x) = \begin{cases} 1 & \text{si } x \geq a, \\ a & \text{si } x < a, \end{cases}, \quad w_a(x) = \begin{cases} 1 & \text{si } x > a, \\ a & \text{si } x \leq a. \end{cases}$$

- (3) If  $F \subseteq A$  is a filter, then  $F$  is always admissible and has the form

$$F = \nabla(w_a) = (a, 1] \quad \text{o} \quad F = \nabla(v_a) = [a, 1]$$

for some  $a \in A$ . In particular,  $F \in A^\wedge$  if and only if  $F = (a, 1]$ .

- (4) If  $F \in A^\wedge$ , the adjusted nucleus associated  $v_F$  is calculated as

$$(15) \quad v_F(x) = l_m(x) = \begin{cases} 1 & \text{if } x > m, \\ x & \text{if } x \leq m. \end{cases}$$

where  $l_m = v_m \wedge w_m$  and  $m = \bigwedge F$ .

- (5) For  $S = \text{pt } A$  it holds that  $S = [0, 1)$  and the open sets in  $\mathcal{O}S$  are of the form

$$\mathcal{U}_A(x) = \{p \in S \mid p < x\} = [0, x).$$

- (6) The frame  $A$  is not  $T_1$ , since no  $p \in S$  is maximal. Consequently,  $A$  does not satisfy any stronger separation property in  $\text{Frm}$ .  
 (7) For  $F \in A^\wedge$  the compact saturated set associated  $Q \in \mathcal{Q}S$  is of the form  $Q = [0, m]$ .  
 (8) It holds that  $[Q'] = v_F$ ; therefore,  $S$  is strongly stacked.  
 (9) Since  $[0, x) \in \mathcal{O}S$ , then  $[x, 1] \in \mathcal{C}S$  and  $Q = [0, m]$ . Therefore, the compact subsets are not closed; that is,  $S$  is not packed.  
 (10) The frame  $A$  is not efficient.

The above resume some characteristics about the frame  $[0, 1]$ . Now, we will perform the analysis of the patch constructions of  $A$  and its point space  $S$ .

Remember that for  $A \in \text{Frm}$  and  $S \in \text{Top}$  we have

$$\text{Pbase} = \{u_a \wedge v_F \mid a \in A, F \in A^\wedge\} \quad \text{y} \quad \text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}$$

where  $\text{Pbase}$  generates the patch frame of  $A$  and  $\text{pbase}$  generates the topology of the patch space of  $S$ . By (15) we have that

$$\text{Pbase} = \{u_a \wedge l_m \mid a \in [0, 1], m \in [0, 1)\}.$$

For  $x \in A$  we have

$$(u_a \wedge l_m)(x) = u_a(x) \wedge l_m(x).$$

Let us analyze the value of this map according to the relative position of  $x$  and  $m$ .

**Case 1:** If  $x < m$ , then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge x = x.$$

**Case 2:** If  $x = m$ , then

$$(u_a \wedge l_m)(x) = u_a(m) \wedge m = x.$$

**Case 3:** If  $x > m$ , then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge 1 = u_a(x).$$

In the first two cases, the value of  $a$  does not affect the evaluation. Thus, the only non-trivial case occurs when  $m < x$ , where the value of  $(u_a \wedge l_m)(x)$  depends on the relative position of  $a$  and  $x$ . We therefore distinguish the following two subcases.

**Case 3.1:** If  $m < x < a$ , then

$$(u_a \wedge l_m)(x) = a.$$

**Case 3.2:** If  $m < a \leq x$ , then

$$(u_a \wedge l_m)(x) = x.$$

From the preceding analysis, we conclude that the non-trivial generators of  $P$ base correspond precisely to the pairs  $(m, a)$  with  $m < a$ . Consequently, the frame  $PA$  is determined by the open intervals of the form  $(m, a)$ , where  $m \in \text{pt } A$  and  $a \in A$ .

We now turn to the question of whether  $PA$  satisfies any separation property. To this end, we recall the relationship between the point-free patch construction and its spatial counterpart.

Let  $A$  be an arbitrary frame and let  $S = \text{pt } A$  be its space of points. The two patch constructions are related by the following commutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & PA & \hookrightarrow & NA \\ \mathcal{U}_A \downarrow & & \downarrow P(\mathcal{U}_A) & & \downarrow N(\mathcal{U}_A) \\ \mathcal{O}S & \longrightarrow & POS & \hookrightarrow & NOS \\ & \searrow & \downarrow \pi & & \downarrow \sigma \\ & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S \end{array}$$

where  $\mathcal{O}^f S$  is the Skula topology (or front topology) of  $S$ . In [8] is named *the patch assembly diagram* and more details about it can also be found in [?].

In our case,  $\mathcal{U}_A$  is an isomorphism. Moreover, under  $\mathcal{U}_A$ , the patch construction turns out to be functorial and, therefore,  $PA \cong POS$ . Similarly,  $NA \cong NOS$ . To describe explicitly  $POS$  and  $NOS$ , we use the following result, which can be consulted in [1].

**Proposition 7.11.** *Let  $S$  be a partially ordered set.*

- (1) *If  $S$  has no infinite antichains, then  $NOS$  is spatial.*
- (2) *If  $S$  is totally ordered, then  $NOS$  is spatial.*

Thus, by Proposition 7.11 we have that  $N\mathcal{O}S$  is spatial and therefore  $N\mathcal{O}S \cong \mathcal{O}^f S$ . With all this information, the patch assembly diagram, in our particular case is as follows.

$$\begin{array}{ccccc} A & \longrightarrow & PA & \longrightarrow & NA \\ \cong \downarrow & & \downarrow \cong & & \downarrow \cong \\ \mathcal{O}S & \longrightarrow & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

Therefore, if we want to study the frame  $PA$ , we can study the patch topology  $\mathcal{O}^p S$ . Remember that

$$\text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}.$$

In our case,

$$\text{pbase} = \{[0, a) \cap [m, 1) \mid a \in [0, 1], m \in [0, 1)\} = \{(m, a)\}.$$

Since the topology generated by the intervals  $(m, a)$  coincides with the usual topology on  $[0, 1)$ , the space  ${}^p S$  is Hausdorff. Consequently,  $\mathcal{O}^p S \cong PA$  satisfies  $(\mathbf{H})$ , even when  $A$  is not  $T_1$  nor neat.

**Lemma 7.12.** *Let  $A$  be a frame, if the morphism  $\text{p}U: \text{p}A \rightarrow \text{p}\mathcal{O}S$  is an isomorphism, then  $A$  is strongly stacked.*

*Proof.* Consider any open filter  $F$  on  $A$  and the compact saturated determined by it, say  $Q$ , thus we have the nuclei  $v_F$  and  $U_*[Q']U = Q[U]$ , we need to observe the action of  $\text{p}U$  on these two nuclei, in order to do that remember that we have a commutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & \text{p}A & \xhookrightarrow{\quad \iota \quad} & NA \\ \mathcal{U}_A \downarrow & & \downarrow \text{p}(\mathcal{U}_A) & & \downarrow N(\mathcal{U}_A) \\ \mathcal{O}S & \longrightarrow & \text{p}\mathcal{O}S & \xhookrightarrow{\quad \iota \quad} & N\mathcal{O}S \\ & \searrow & \downarrow \pi & & \downarrow \sigma \\ & & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

since  $N(\mathcal{U})\iota = \text{p}\mathcal{U}$ , that is,  $\text{p}\mathcal{U}$  is the restriction of  $N(\mathcal{U})$  to  $\text{p}A$ , then

$$\text{p}\mathcal{U}(v_F) = N(\mathcal{U})(v_F) = \bigvee \{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

since the morphism  $\mathcal{U}$  transforms open filters we have

$$\bigvee \{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\} = \bigvee \{u_{v_{\mathcal{U}F}} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

But this representation is the corresponding to the nucleus  $v_{\mathcal{U}F}$ .

□

Completar esta prueba

Considerar incluir información sobre los parches